

Resolving a Conjecture on Stack-Sorting

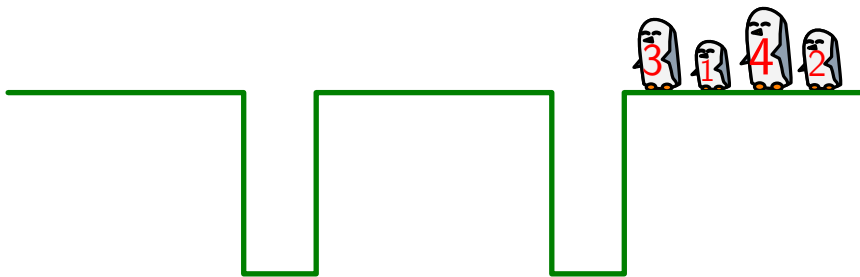
Michael Luo
Research Science Institute
Mentor: Ryota Inagaki

August 2026

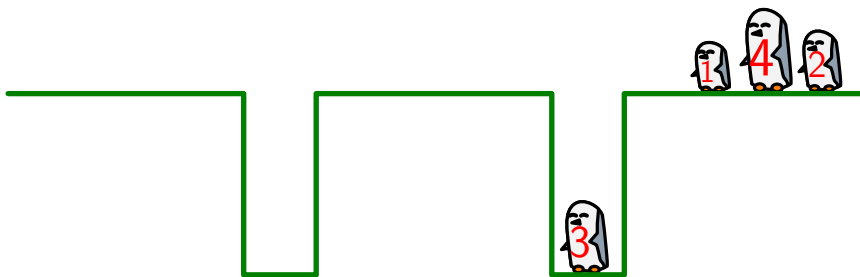
Penguins

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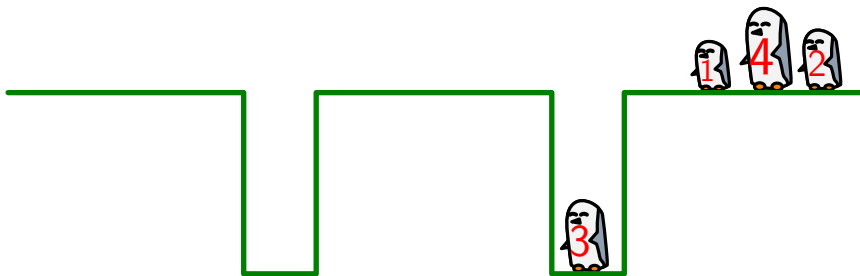


Penguins by Evan Chen, used with permission. Figure generated with AI assistance.



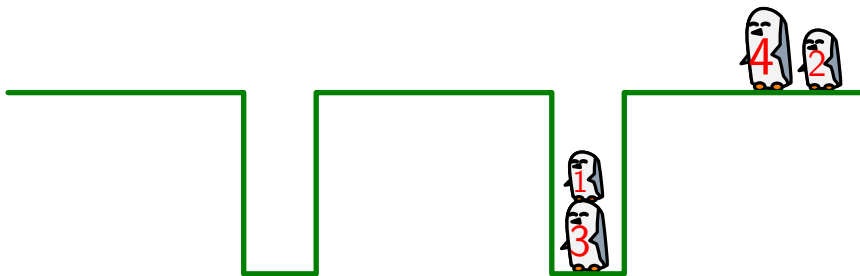
Penguins are

- **Impatient:** small penguins will always jump on big penguins



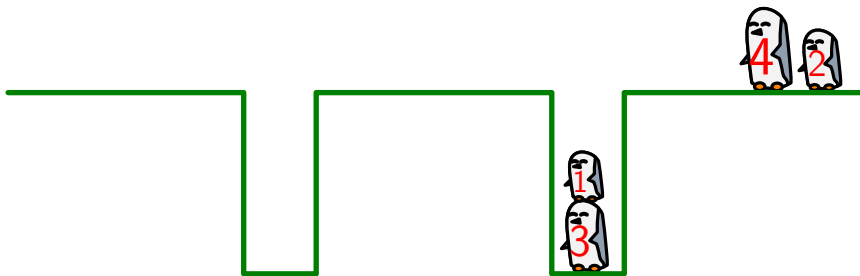
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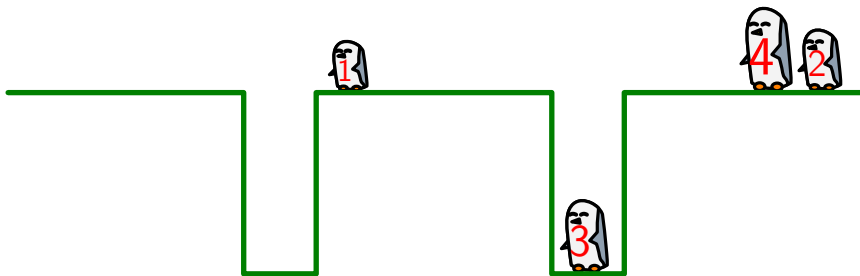
Penguins are

- **Impatient:** small penguins will always jump on big penguins
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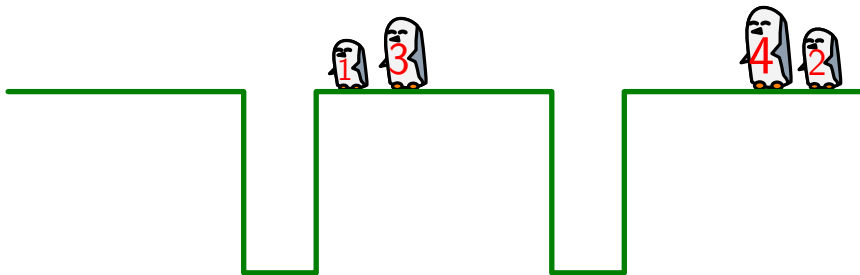
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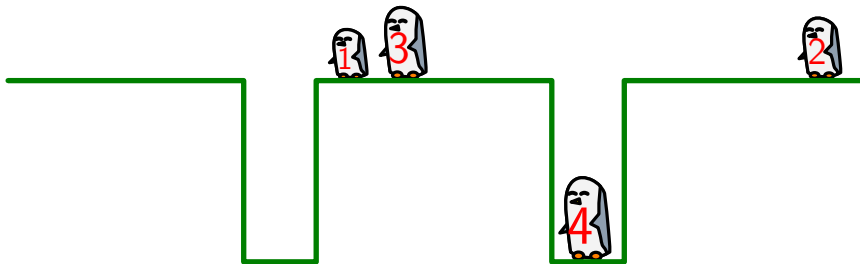
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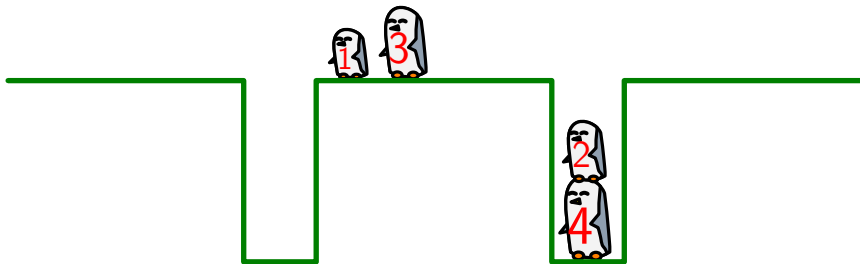
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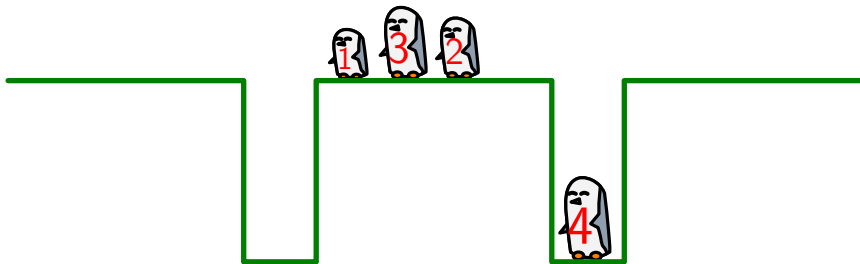
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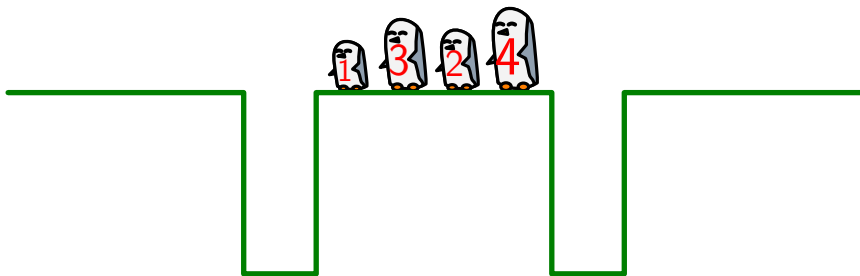
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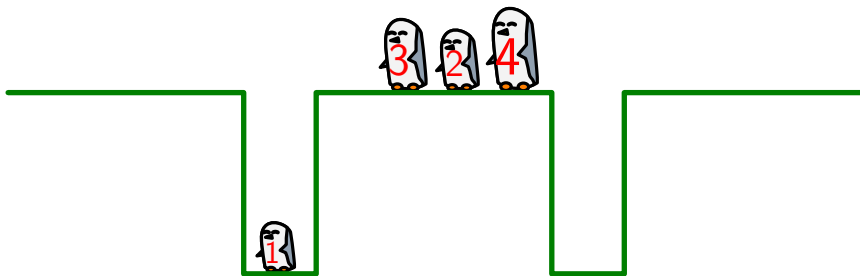
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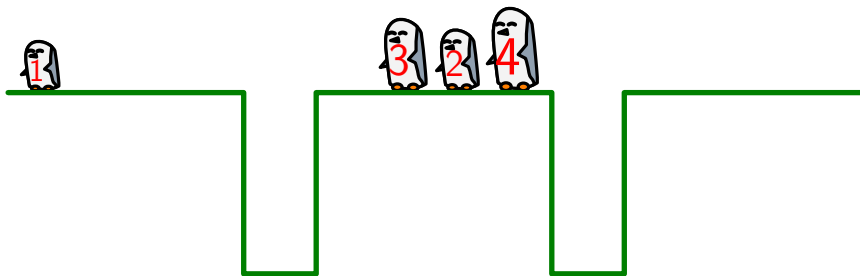
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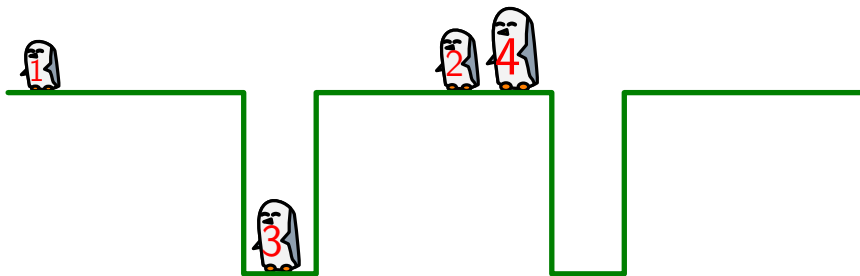
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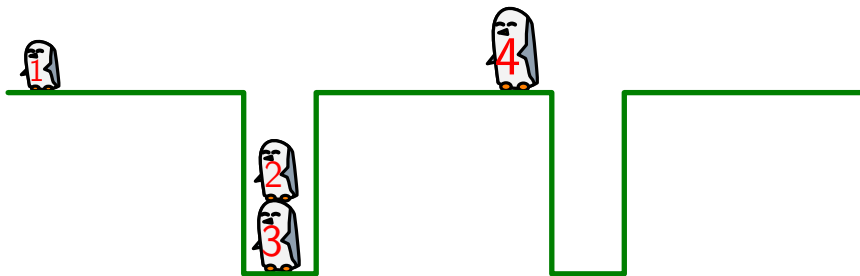
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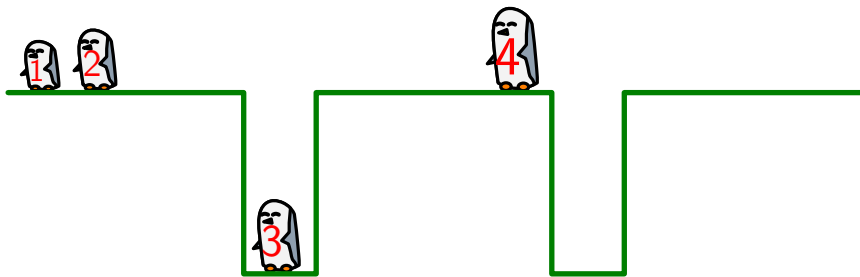
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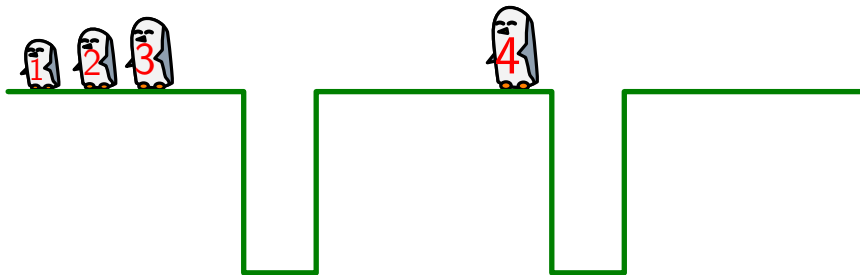
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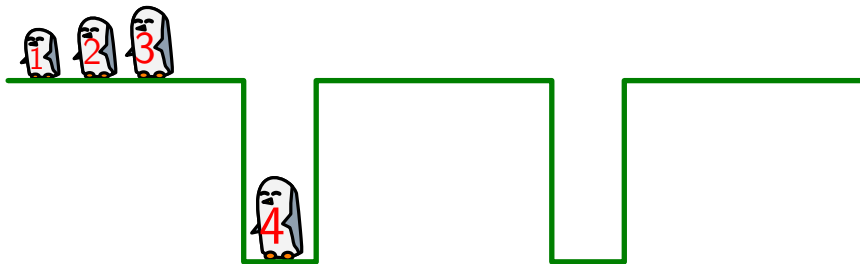
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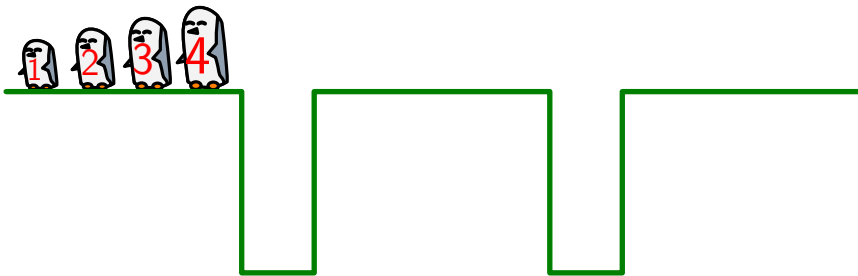
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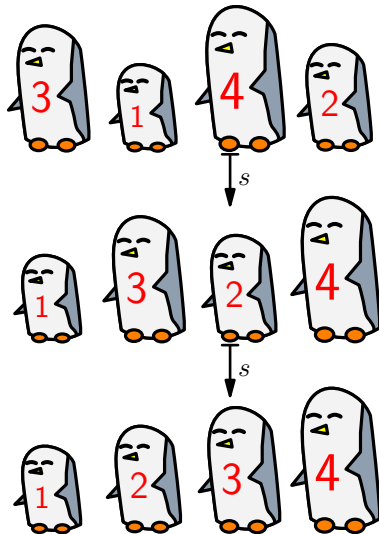


Penguins by Evan Chen, used with permission. Figure generated with AI assistance.

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In notation: $s(3142) = 1324$, $s(1324) = 1234$

Main result

$$S_3 = \{123, 132, 213, 231, 312, 321\}$$

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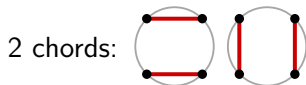
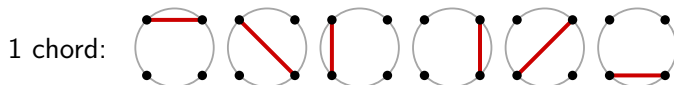
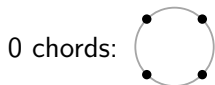
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Theorem (L. 2026)

of s^2 -sortable permutations in S'_n is n th Motzkin number.

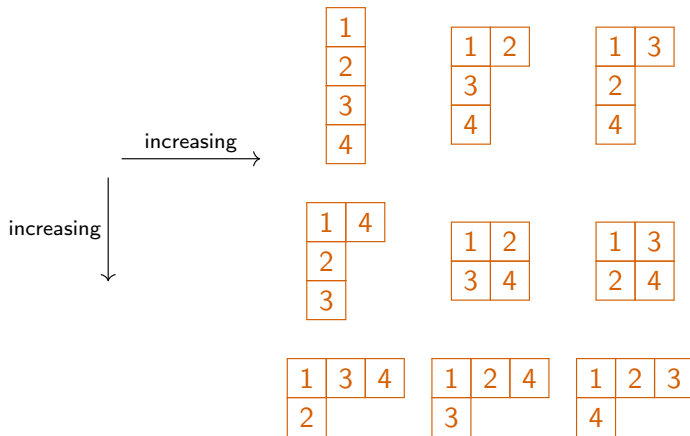
n th Motzkin counts:

Ways to draw any # of disjoint chords connecting n points on a circle



n th Motzkin counts:

Number of **standard Young tableaux** with width ≤ 3 (Regev 1981)



*n*th Motzkin counts:

Number of 4321-, {3412, 2413}, {3412, 3142}- and 3412-avoiding involutions in S_n .
 Number of sequences of length $n-1$ consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any 2 consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003
 From [David Callan](#), Jul 15 2004: (Start)
 Also number of Motzkin n -paths: paths from $(0,0)$ to $(n,0)$ in an $n \times n$ grid using only steps $U = (1,1)$, $F = (1,0)$ and $D = (1,-1)$.

Number of Dyck n -paths with no UUU. (Given such a Dyck n -path, change each UUD to U, then change each remaining UD to F. This is a bijection to Motzkin n -paths. Example with $n=5$: U U D U D U U D D D $\rightarrow$ U F U D D.)

Number of Dyck $(n+1)$ -pathsthz no UDU. (Given such a Dyck $(n+1)$ -path, mark each $\downarrow$ that is followed by a D and each D that is not followed by a U. Then change each unmarked U into which matching D is marked to an F. Lastly, delete all the marked steps. This is a bijection to Motzkin n -paths. Example with $m=6$ and marked steps shown in red:

$a(n)$: the number of strings of length $2n+2$ from the following recursively defined set: L contains the empty string and, for any strings a and b in L, we also find (ab) in L. The first few elements of L are e , i , $((i))$, $((i))i$, $((i))((i))$, $((i))((i))i$, $((i))((i))i$, $((i))((i))i$ and so on. This proves that $a(n)$ is less than or equal to $C(n)$, the n -th Catalan number. See Orrick Link (2024). – Saul Schleimer

[sausch@math.rutgers.edu], Feb 23 2006 [Additional linked content added by William P. Orrick, Jun 13 2024]

$a(n)$ = number of Dyck n -paths all of whose valleys have even x -coordinate (when path starts at origin). For example, $T(4,2)=3$ counts UDUUDD, UDUUDDU, UDUUDDU. Given such a path, split it into n subpaths of length 2 and transform UU $\rightarrow$ U, DD $\rightarrow$ D, UD $\rightarrow$ F (there will be no DFs for that would entail a valley with odd x -coordinate). This is a bijection to Motzkin n -paths. — David Callan, Jun 07 2006

Also the number of standard Young tableaux of height $k = 3$. - Mike Zabrocki, Mar 24 2007
 a(n) is the number of RNA shapes of size $2n+2$. RNA Shapes are essentially Dyck words without "directly nested" motifs of the form A[B]C, for A, B and C Dyck words. The first RNA Shapes are []; [][]; {}{}; {}{}{}; {}{}{}{}; {}{}{}{}{}; {}{}{}{}{}{}; ... - Yann Ponty (ponty@ATRI.fr), May 30 2007
 The sequence is self-generated from top row A going to the left starting (1,1) and bottom row = 8, the same as the previous one, and then moving up and right until you reach the end of the row. Then move down and repeat the sequence A to get the next step of A. Example: A[5] = 23, 6, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 80

1, 1) and $\{0, 1, 1, 2, 4\} = \{0, +4+2+2+4\} = 12$; which is added to $9 = 21$. - Gary W. Adamson, Oct 27 2008

Equals $A005773 + A005773$ shifted (i.e., $\{1, 2, 5, 13, 35, 96, \dots\} / \{1, 1, 2, 5, 13, 35, 96, \dots\}$). - Gary W. Adamson, Dec 21 2008

Starting with offset 1 = states of $M = [1,1,0,0,0, \dots]$, where M = a tridigital matrix with $[0,1,1,1, \dots]$ in the main diagonal and $[1,1,1, \dots]$ in the super and subdiagonals. Gary W. Adarson, Jan 07 2089
 $a(n)$ is the number of involutions of $\{1,2, \dots, n\}$ having genus 0. The genus $g(p)$ of a permutation $p = [1,2, \dots, n]$ is defined by $g(p) = (1/2)(n - (p) - \text{cyc}(p))$, where $p =$ the inverse image of p , $(p) =$ the number of cycles in p , and $\text{cyc}(p) =$ the number of cycles in p . Example: $a(4) = 4$, indeed, $p = 3412 = (32)(24)$ is the only involution of $\{1,2,3,4\}$ with genus > 0 . This follows easily from the fact that a permutation p of $\{1,2, \dots, n\}$ has genus 0 if and only if the cycle decomposition of p gives a noncrossing partition of $\{1,2, \dots, n\}$ and each cycle of p is increasing (see Lemma 2.1 of the Bulwicz-Simion reference).
 ERIC F. DEUSCH, May 29 2010

Let $w(i, j, n)$ denote walks in $\mathbb{N}^2$ which satisfy the multivariate recurrence $w(i, j, n) = w(i, j+1, n-1) + w(i-1, j, n-1) + w(i+1, j-1, n-1)$ with boundary conditions $w(0, 0, 0) = 1$ and $w(i, j, n) = 0$ if i or j or n is < 0 . Then $a(n) = \sum_{i=0, n}^n \sum_{j=0, n}^n w(i, j, n)$ is the number of such walks of length n .

[Peter Luschny](#), May 21 2011
 $a(n)/a(n-1)$ tends to 3.0 as $N \rightarrow \infty$: $(1+2\cos(2\pi/N))$ relating to longest odd N regular polygon diagonals, by way of example, $N=7$: Using the tridiagonal generator [cf. comment of Jan 07 2009], for

polygon N=7, we extract an $(N-1)/2 = 3 \times 3$ matrix, $[0,1,0; 1,1,1; 0,1,1]$ with an e-val of 2.24697...; the longest Heptagon diagonal with edge = 1. As N tends to infinity, the diagonal lengths tend to 3.0, the convergent of the sequence. - Gary W. Adanson, Jun 08 2011

Number of $(n+1)$ -length permutations avoiding the pattern 132 and the dotted pattern $23\dot{\{1\}}$. - Jean-Luc Baril, Mar 07 2012

Number of n -length words w over alphabet $\{a,b,c\}$ such that for every prefix z of w we have $\#(z,a) \geq \#(z,b)$

$\geq \#(z, c)$, where $\#(z, x)$ counts the letters x in word z . The $a(4) = 9$ words are: aaaa, aaab, aaba, abaa, aabb, abab, aabc, abac, abca. — [Alois P. Heinz](#), May 26 2012

Number of length- n restricted growth strings (RGS) $[r(1), r(2), \dots, r(n)]$ such that $r(1)=1$, $r(k) \leq k$, and

`r[k] := r[k-1]`; for example, the 9 RGS for $n=4$ are 1010, 1012, 1201, 1210, 1212, 1230, 1231, 1232, 1234. -
Joerg Arndt, Apr 16 2013

Number of length- n restricted growth strings (RGS) $[r(1), r(2), \dots, r(n)]$ such that $r(1)=0$, $r(k) \leq k$ and

Number of (4231.5276143)-avoiding involutions in S_n . - [Alexander Burstein](#), Mar 05 2014

$a(n)$ is the number of increasing unary-binary trees with n nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A245888](#). - [Manda Riehl](#), Aug 07 2014

From [Tony Foster III](#), Jul 28 2016: (Start)
A series created using $2a(n) + a(n+1)$ has Hankel transform of $F(2n)$, offset 3, F being the Fibonacci bisection. [A081986](#) (empirical observation).

A series created using $2a(n) + 3a(n+1) + a(n+2)$ gives the Hankel transform of $\text{Sum}_{k=0..n} k! \text{Fibonacci}(2k)$, offset 3, [A197649](#) (empirical observation). (End)

Conjecture: $(2n+1) + \sum_{k=1}^n (2k+1) + a(n)$, 2 is an integer for each positive integer n . [2017-06-03](#), Nov 16 2017

The Rubey and Stump reference proves a refinement of a conjecture of René Marczinkin, which they state as:
 "If the cube of a 2, 3, 4, or 5 sided plane or solid non-Mohrman 2, 3, 4, or 5 sided polyhedron and their co-

The number of 2-Gorenstein algebras which are Wakayama algebras with n simple modules and have an oriented line as associated quiver equals the number of Motzkin paths of length n . - [Eric M. Schmidt](#), Dec 16 2017

number of $U_{\tau_1, \tau_2}(k)$ -equivalence classes of Lukasiewicz paths. Lukasiewicz paths are P -equivalent iff the positions of pattern P are identical in these paths. - [Sergey Kirgizov](#), Apr 08 2018

If τ_1 and τ_2 are two distinct permutation patterns chosen from the set $\{132, 231, 312\}$, then $a(n)$ is the

number of valid hook configurations or permutations of $[n+1]$ that avoid the patterns τ_{au_1} and τ_{au_2} . -
[Colin Defant](#), Apr 28 2019
 Number of permutations of length n that are sorted to the identity by a consecutive-321-avoiding stack

From [Helmut Prodinger](#), Dec 13 2020: (Start)
 $a(n)$ is the number of paths in the first quadrant starting at $(0,0)$ and consisting of n steps from the

For example, denoting $U=(1,1)$, $D=(1,-1)$, $D_{-j}=(1,-j)$ for $j \geq 2$, $a(4)$ counts $UUUU$, $UUUD$, $UUUD_2$, $UUUD_3$, $UUUD_4$, $UUUD_5$, $UUUD_{-2}$, $UUUD_{-3}$, $UUUD_{-4}$, $UUUD_{-5}$.

This step set is inspired by $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$, suggested by Emeric Deutsch around 2000. See [Prodinger link](#) that contains a bijection to Motzkin paths. (End)

Named by Donaghey (1977) after the Israeli-American mathematician Theodore Motzkin (1908–1970). In Sloane's

"A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". - [Amiram Eldar](#), Apr 15 2021

number of odd Motzkin n -paths. The value $A107507(n) - A343386(n)$ can be called the "shadow" of $a(n)$ (see [A343773](#)). – Gennady Eremin, May 17 2021

Conjecture: If p is a prime of the form $6n+1$ ([A002476](#)), then $a(p-2)$ is divisible by p . Currently, no

counterexample exists for $p < 10^7$. Personal communication from [Robert Gerbicz](#): mod such p this is equivalent to [A866795](#) with comment: "Every [A866795](#)(n) from [A866795](#)((p-1)/2) to [A866795](#)(p-1) is divisible by prime p of form $6n+1$ ". - [Serge Batalov](#), Feb 08 2022

The conjecture is proved in the 2017 paper by Rob Burns in the Links below. The result is contained in Tables 4 and 5 of the paper, which show that $a(p-2) \equiv 0 \pmod{p}$ when $p \equiv 1 \pmod{6}$ and $a(p-2) \equiv -1 \pmod{p}$ when $p \equiv 5 \pmod{6}$.

In fact, the 2017 paper by Burns establishes more general congruences for $a(p^k - 2)$ where $k \gg 1$. If $p \equiv 1 \pmod 6$ then $a(p^k - 2) \equiv 0 \pmod n$ for $k \gg 1$.

If $p \equiv -1 \pmod{6}$ then $a(p^k - 2) \equiv -1 \pmod{p}$ when k is odd and $a(p^k - 2) \equiv 0 \pmod{p}$ when k is even. These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper. The 2024 paper by Nadav Kaban also proves the conjecture. Proposition 6 of the paper states that a prime p

From [Peter Bala](#), Feb 10 2022: (Start)
 Conjecture: If p is a prime, then p divides $a(p-2)$ if and only if $p \equiv 1 \pmod{3}$. (End)

(1) For prime $p \equiv 1 \pmod{6}$ and $n, r \geq 1$, $a(n+p^r-2) \equiv -A085717(n-1) \pmod{p}$, where we take $A085717(0) = 0$ to match Batalov's conjecture above.

(4) For prime $p \geq 5$ and $k \geq 2$, $a(n + p^k) \equiv a(n) \pmod{p^2}$ for $0 \leq n \leq \{p^k(k-1) - 3\}$. (End)

the Hankel transform of the sequence with a(0) omitted gives the period-6 sequence [1, 0, -1, -1, 0, 1, ...] which is [A018892](#) with its first term omitted, while the Hankel transform of the current sequence is the all-ones sequence [A000012](#), and also it is the unique sequence with this property which is similar to

The number of terms in which the exponent of any variable x_i is not greater than 2 in the expansion of $\text{Product}_{j=1..n} \text{Sum}_{i=1..j} x_i$. E.g.: $a(4) = 9: 3x_1^2x_2x_3^2, 4x_1^2x_2x_3x_4, 2x_1^2x_2x_4^2, x_1^2x_3^2,$

The Hankel matrix corresponding to this sequence factorizes as $T \cdot \text{Transpose}(T)$, where T is the lower unitriangular array [A064189](#) (row reverse of the Motzkin triangle [A026300](#)). Hence the Hankel determinants

are all equal to 1 as stated above by Michael Somos. - [Peter Bala](#), Nov 23 2025
 From [Michael Luo](#), Jul 29 2026: (Start)
 Number of 2-stack-sortable permutations of $\{1, 2, \dots, n+1\}$ ending with 1.

Number of standard composition tableaux of width at most 2 on n cells. (End)

Theorem (L. 2026)

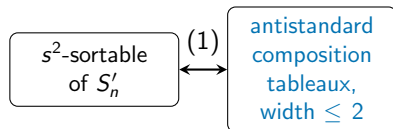
of s^2 -sortable permutations in S'_n is n th Motzkin number.

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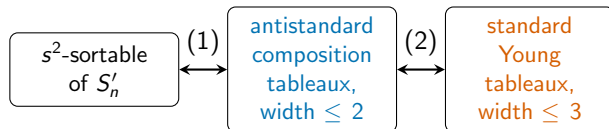
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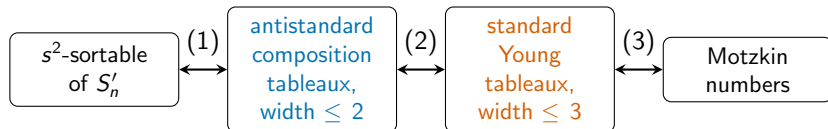


Proof idea

Theorem (L. 2026)

of s^2 -sortable permutations in S'_n is n th Motzkin number.

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Lemma (Chen–L.–Zhang 2025, Zhang 2026)

Take $\pi \in S'_n$ and apply s repeatedly. Then

- each time, the *right-to-left maxima* of the part before 0 go after 0 and
- the part after 0 is always sorted.

$$4\mathbf{6}521\mathbf{3}0 \xrightarrow{s} 412\mathbf{0}356$$

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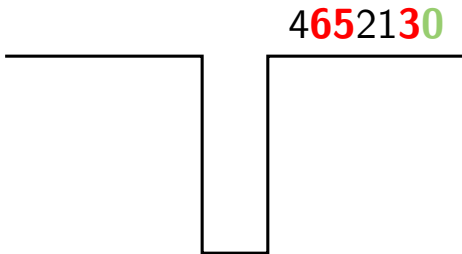
$$\mathbf{1}023456 \xrightarrow{s} 0\mathbf{1}23456$$

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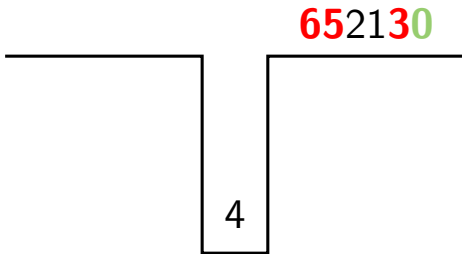


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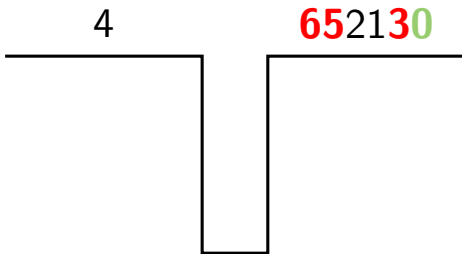


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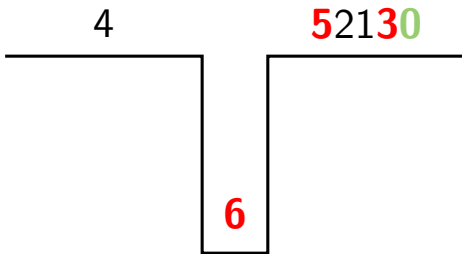


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$$4652130 \xrightarrow{s} 4120356$$

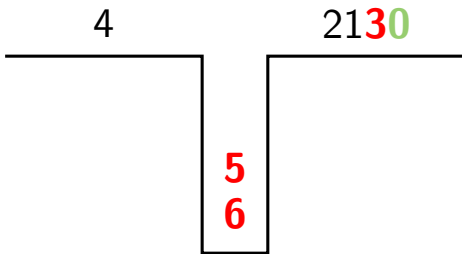


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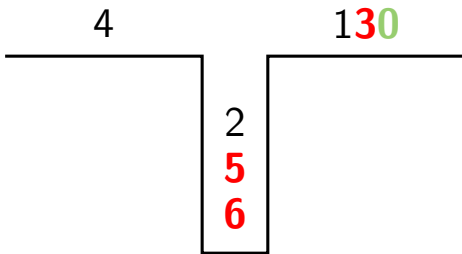


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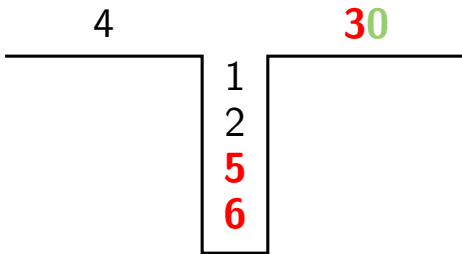


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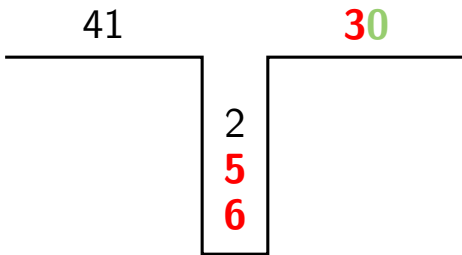


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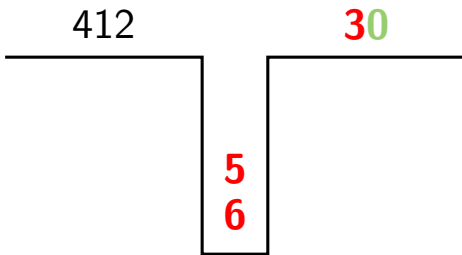


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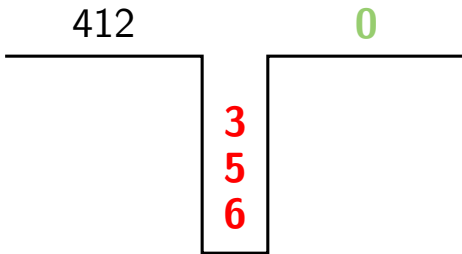


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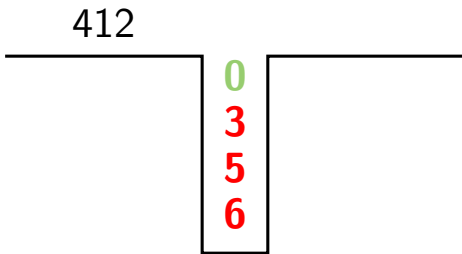


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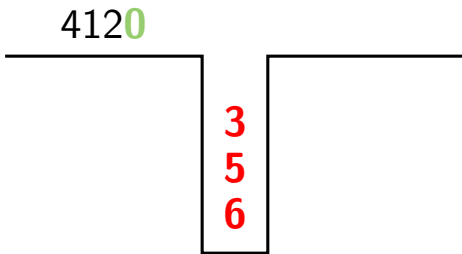


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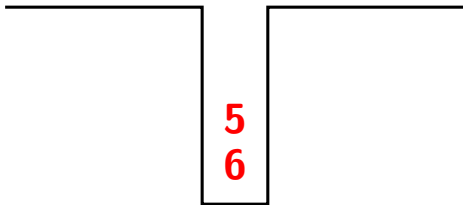
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41203



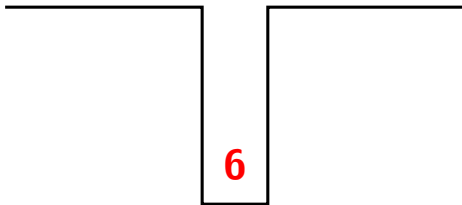
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$$4\mathbf{6}521\mathbf{3}0 \xrightarrow{s} 412\mathbf{0}35\mathbf{6}$$

412035



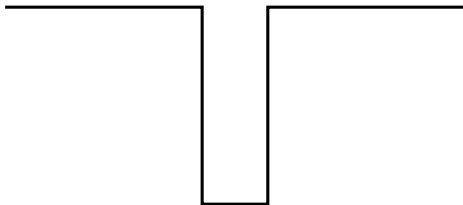
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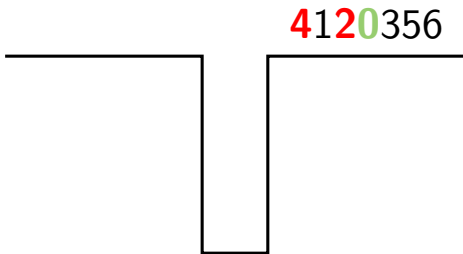


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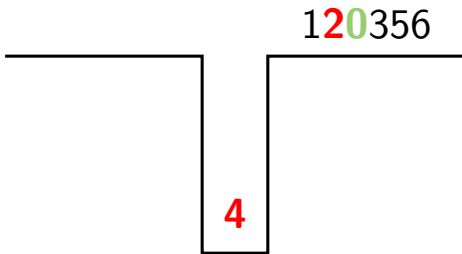


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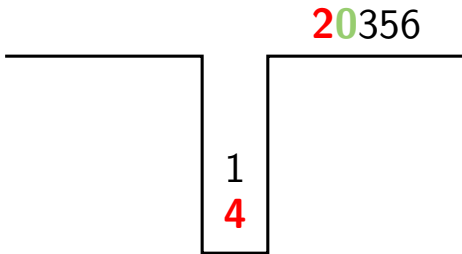


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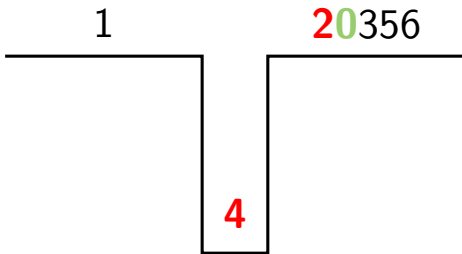


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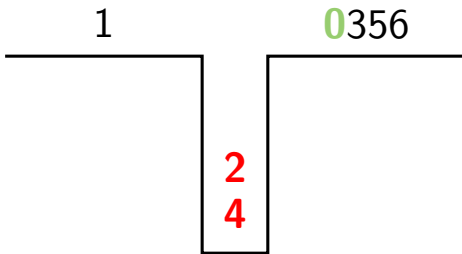


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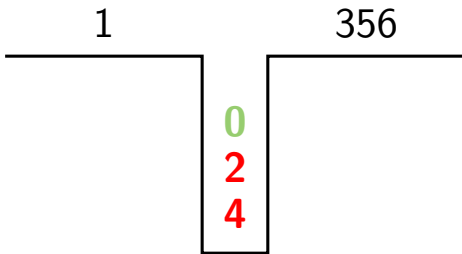


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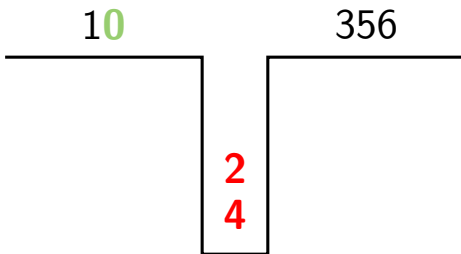


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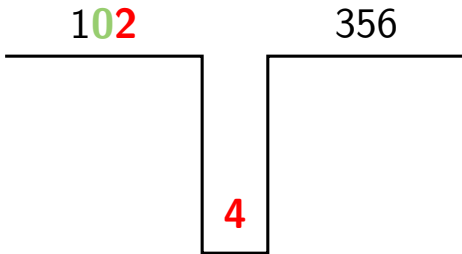


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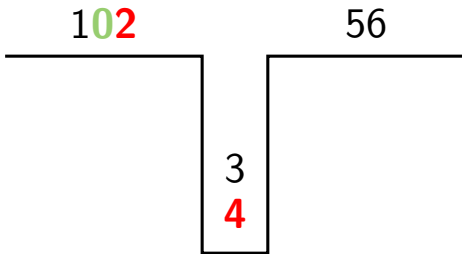


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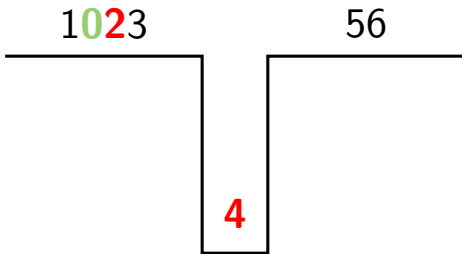


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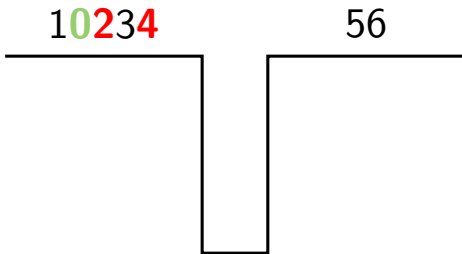


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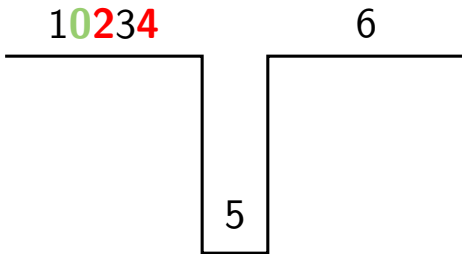


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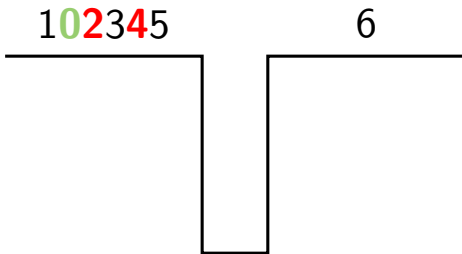


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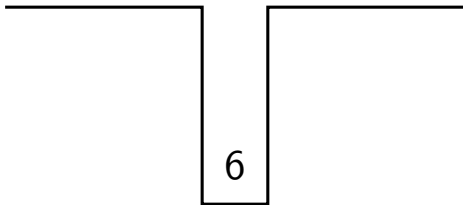
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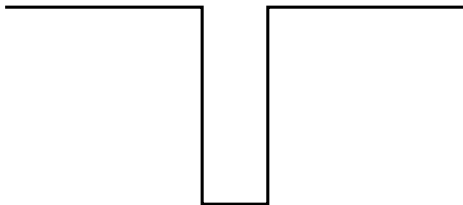
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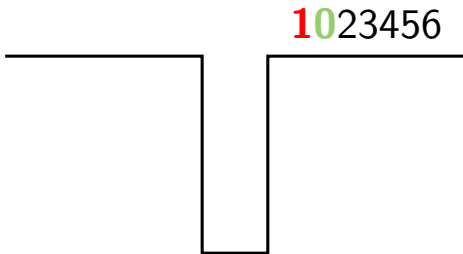


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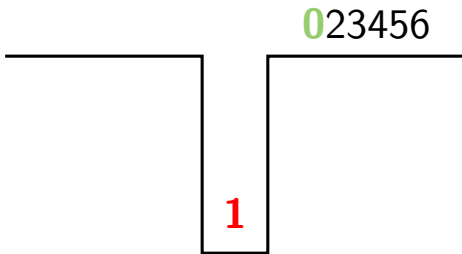


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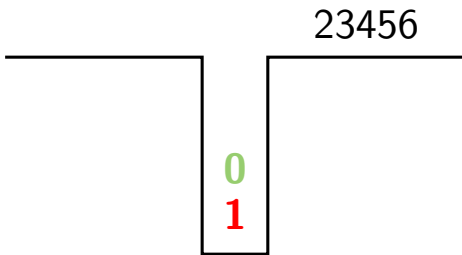


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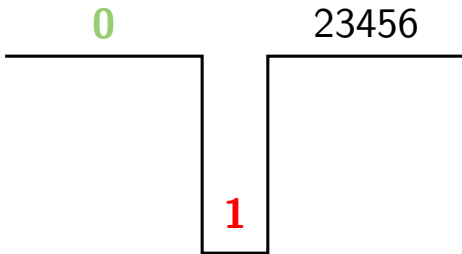


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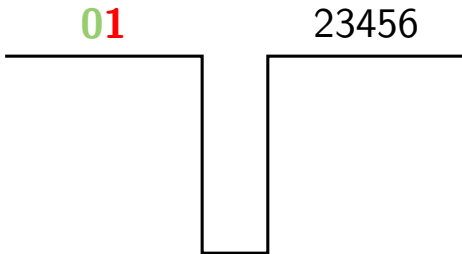


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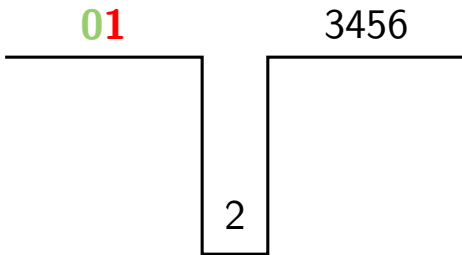


Lemma (Chen–L.–Zhang 2025, Zhang 2026)

Take $\pi \in S'_n$ and apply s repeatedly. Then

- each time, the *right-to-left maxima* of the part before 0 go after 0 and
- the part after 0 is always sorted.

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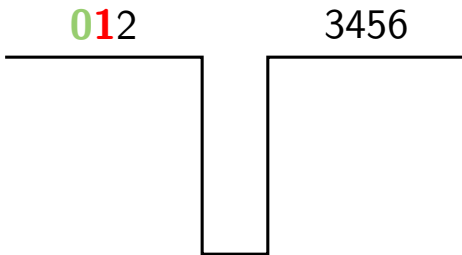


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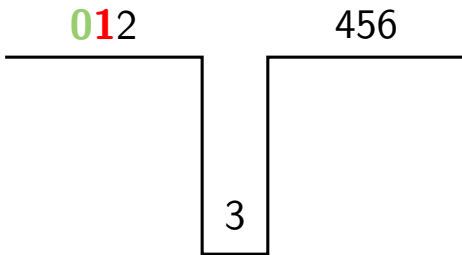


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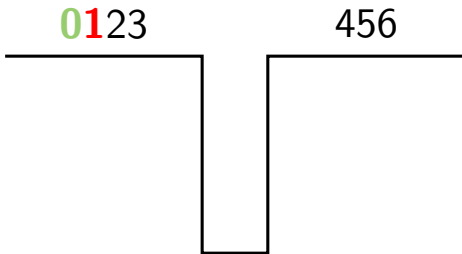


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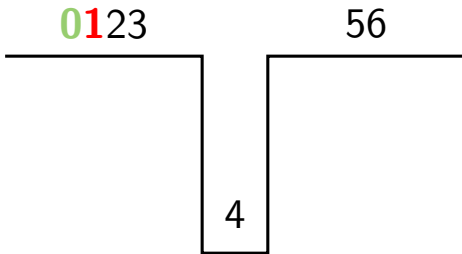


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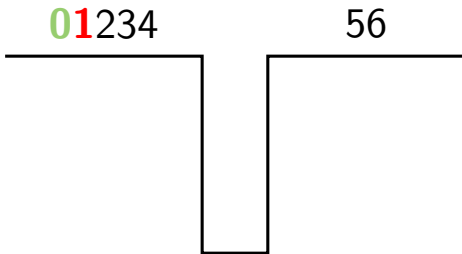


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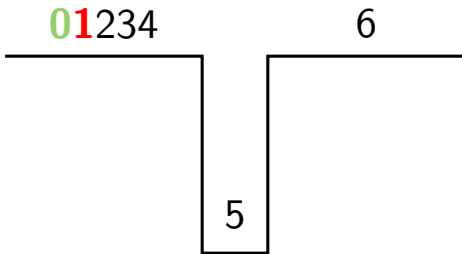


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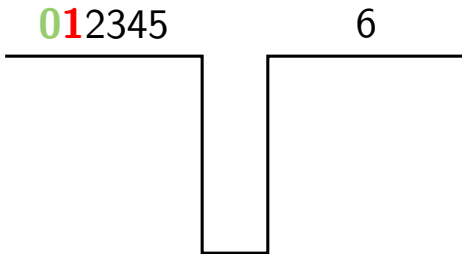


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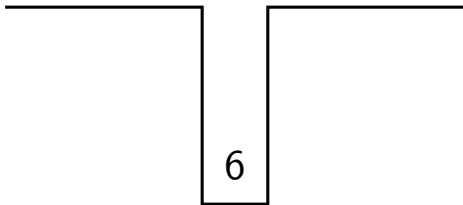
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$$1023456 \xrightarrow{s} 0123456$$

012345



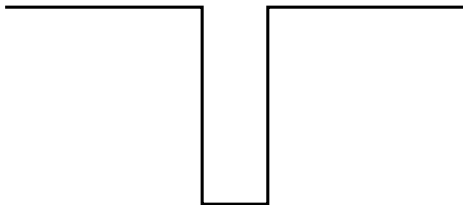
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0123456



Lemma (Chen–L.–Zhang 2025, Zhang 2026)

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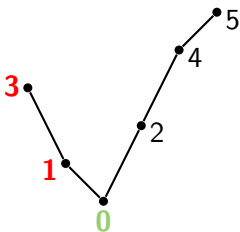
- each time, the *right-to-left maxima* of the part before 0 go after 0 and
- the part after 0 is always sorted.

$$4\mathbf{6}521\mathbf{3}0 \xrightarrow{s} 4120\mathbf{3}56$$

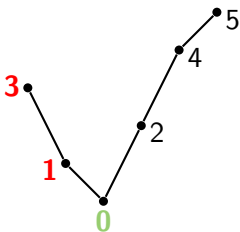
$$\mathbf{4}120\mathbf{3}56 \xrightarrow{s} 10\mathbf{2}3\mathbf{4}56$$

$$\mathbf{1}023456 \xrightarrow{s} 0\mathbf{1}23456$$

310245 is s-sortable since all elements before 0 are right-to-left maxima

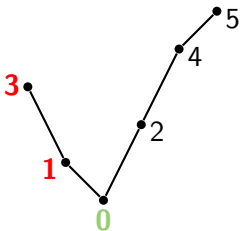


310245 is s -sortable since all elements before 0 are right-to-left maxima

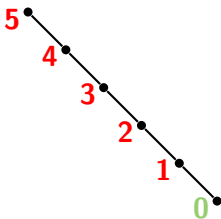


s -sortable permutations are V-shaped!

310245 is s -sortable since all elements before 0 are right-to-left maxima



s -sortable permutations are V-shaped! Only one s -sortable in S'_n :



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

42310

good

42130

bad

If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

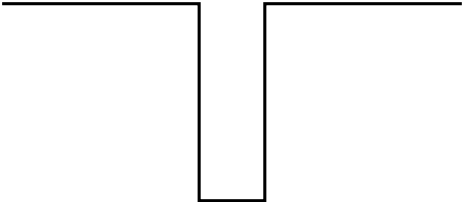
42310

good

42130

bad

42130



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

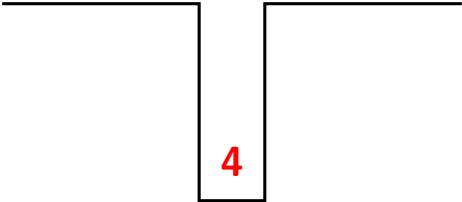
42310

good

42130

bad

2130



4

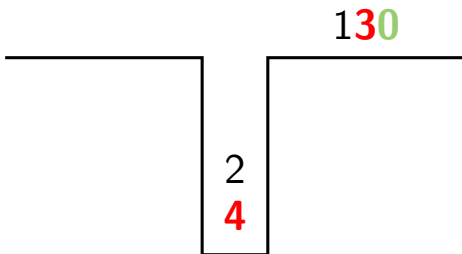
If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

42310

good

42130

bad



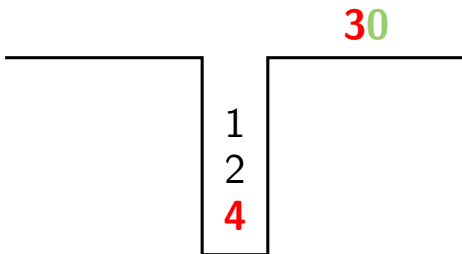
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42310

good

42130

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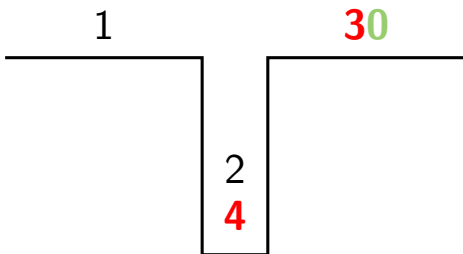
If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

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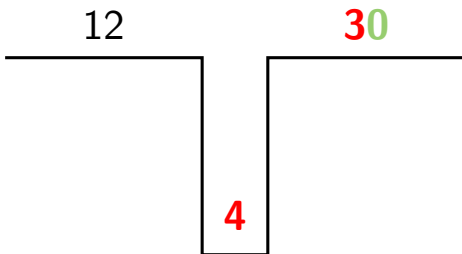
If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

42310

good

42130

bad



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

42310

good

42130

bad

12

0

3

4

If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

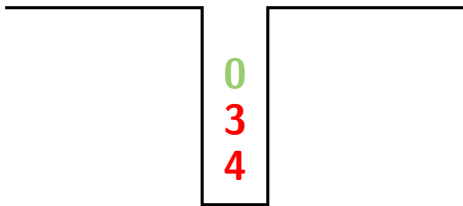
42310

good

42130

bad

12



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

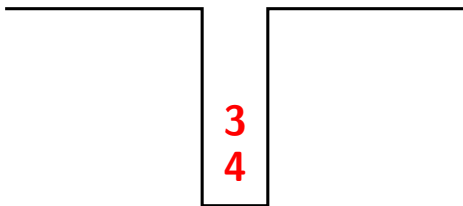
42310

good

42130

bad

120



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

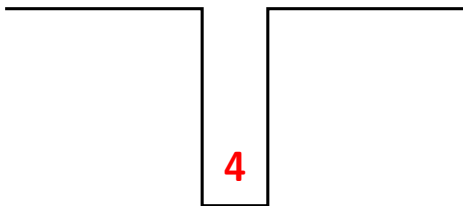
42310

good

42130

bad

1203



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

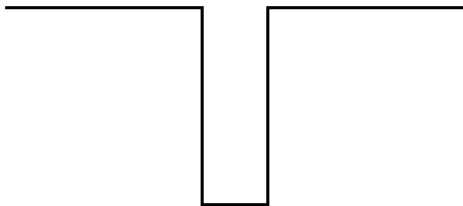
42310

good

42130

bad

12034



354120

487635120

3**5**41**2**0



5	3
4	
2	1

4**8**7**6**3**5**1**2**0



8	4
7	
6	
5	3
2	1

354120

$\longleftrightarrow$

5	3
4	
2	1

487635120

$\longleftrightarrow$

8	4
7	
6	
5	3
2	1

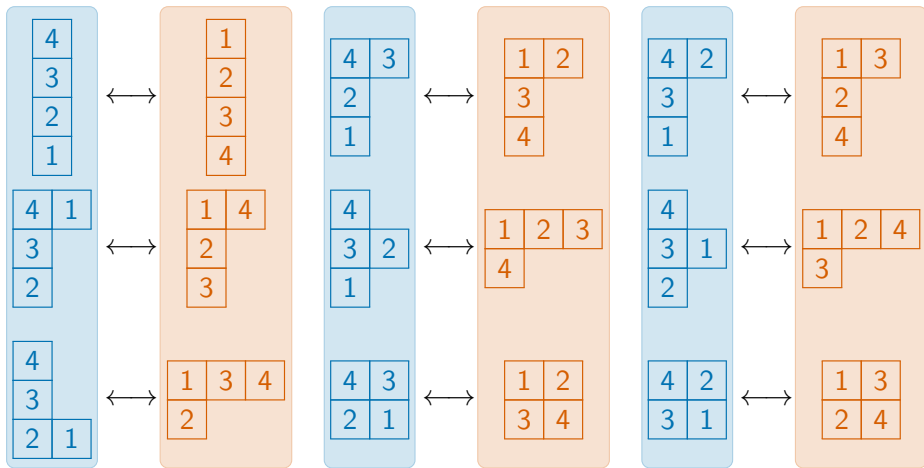
- Rows and columns decreasing
- Entries are exactly $\{1, 2, \dots, n\}$
- Width at most 2

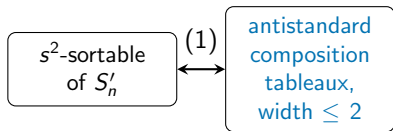
“Antistandard composition tableaux with width at most 2”

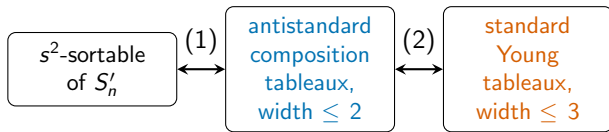
My result:

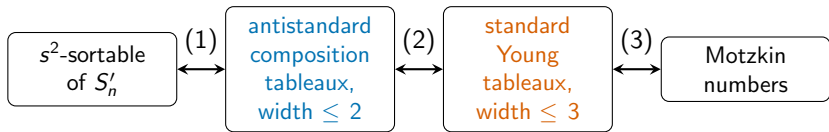
antistandard composition tableaux, width $\leq 2 \longleftrightarrow$

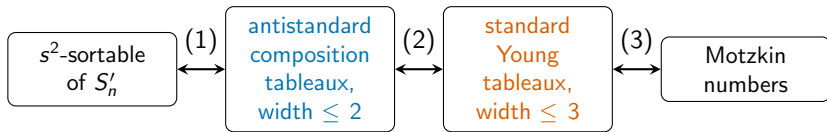
standard Young tableaux, width ≤ 3











Theorem (L. 2026)

of s^2 -sortable permutations in S'_n is n th Motzkin number.

*n*th Motzkin counts:

Number of 4321-, {3412, 2413}, {3412, 3142}- and 3412-avoiding involutions in S_n .
 Number of sequences of length $n-1$ consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any 2 consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003
 From [David Callan](#), Jul 15 2004: (Start)
 Also number of Motzkin n -paths: paths from $(0,0)$ to $(n,0)$ in an $n \times n$ grid using only steps $U = (1,1)$, $F = (1,0)$ and $D = (1,-1)$.

Number of Dyck n -paths with no UUU . (Given such a Dyck n -path, change each UUD to U , then change each remaining UD to F . This is a bijection to Motzkin n -paths. Example with $n=5$: $UUDUDUUDDD \rightarrow UFUUDDD$.)

[illegible]

a(n) = number of Dyck n -paths all of whose valleys have even x -coordinate (when path starts at origin). For example, $T(4,2)=3$ counts UDUUDD, UDUUCDD, UDUUDUD. Given such a path, split it into n subpaths of length 2 and transform UU $\rightarrow$ U, DD $\rightarrow$ D, UD $\rightarrow$ F (there will be no DFs for that would entail a valley with odd x -coordinate). This is a bijection to Motzkin n -paths. - David Callan, Jun 03 2006

Also the number of standard Young tableaux of height $k = 3$. - Mike Zabrocki, Mar 24 2007
 a(n) is the number of RNA shapes of size $2n-2$. RNA Shapes are essentially Dyck words without "directly nested" motifs of the form A[B]B[C], for A, B and C Dyck words. The first RNA Shapes are []; [] [] []; [] [] [] [] []; [] [] [] [] [] []; [] [] [] [] [] [] [].
 The sequence is self-generated from top row A going to the left starting (1,1) and bottom row B; B, the same sequence but starting (0,1) and going to the right. Take dot product of A and B and add the result to n-th term of A to get the (n+1)-th term of A. Example: a(5) = 21 as follows: Take dot product of A = (9, 4, 2, 1, 1), B = (1, 1, 1, 1, 1).

2088
Equals $A055773 / A05773$ shifted (i.e., $(1,2,5,13,35,96,\dots) / (1,1,2,5,13,35,96,\dots)$). - Gary W. Adamson,
Dec 21 2008
Starting with offset 1 = iterates of $M = [1,0,0,0,\dots]$, where M = a tridiagonal matrix with $[0,1,1,\dots]$
in the main diagonal and $[1,1,1,\dots]$ in the super and subdiagonals. - Gary W. Adamson, Jan 07 2009.

$n!$ is the number of involutions of $\{1, 2, \dots, n\}$ having genus g . The genus $g(p)$ of a permutation p of $\{1, 2, \dots, n\}$ is defined by $g(p) = (1/2)(n - 1 - (p - \text{cpl}(p)))$, where p^* is the inverse permutation of p , $c = 234\dots n-1 = \{1, 2, \dots, n\}$, and $\text{cpl}(p)$ is the number of cycles of the permutation p . Example: $a(4) = 9$; indeed, $9 = 1 + 2 + 2 + 2 + 2$ is the sum of the number of involutions of $\{1, 2, \dots, 4\}$ having genus g . Note that a permutation p of $\{1, 2, \dots, n\}$ has genus 0 if and only if the cycle decomposition of p gives an undirected partition of $\{1, 2, \dots, n\}$ and each cycle of p is increasing (see Lemma 2.1 of the Duluc-Sun-Zhang reference). [Also, redundantly, for $p = 3412$ in (13) [24] we have $\text{cpl}(p) = 2341 + 3412 + 2123 = 1432$ and so $g(p) = (1/2)(4 + 1 - 2 - 1) = 1$.

Let $w(i, j, n)$ denote walks in $\mathbb{N}^2$ which satisfy the multivariate recurrence $w(i, j, n) = w(i, j+1, n-1) + w(i-1, j, n-1) + w(i+1, j-1, n-1)$ with boundary conditions $w(0, 0, 0) = 1$ and $w(i, j, n) = 0$ if i or j or n is < 0 . Then $a(n) = \sum_{i=0..n} \sum_{j=0..n} w(i, j, n)$ is the number of such walks of length n . - Peter Luschny, May 21 2011

$\sin(\pi/n-1)$ tends to 3.0 as $N \rightarrow \infty$: $1+2\cos(2\pi/N)$ relating to longest odd N regular polygon diagonals, by way of example, $N=7$: Using the triangular generator [cf. comment of Jan 07 2009], for polygon $N=7$, we extract an $(N-1)/2 = 3 \times 3$ matrix, $\{0,1\}$, $\{0,1,1; 1,1,1; 0,1,1\}$ with an e-val of $2.4697\dots$, for the longest Heptagon diagonal with edge = 1 . As N tends to infinity, the diagonal lengths tend to 3.0 , the convergent of the sequence. — Gary W. Adamson, Jun 08 2011
 N-length permutations avoiding the pattern 132 and the dotted pattern 23d{ot}1}. — Jean-Luc Baril, Mar 07 2012

Number of n -length words w over alphabet $\{a,b,c\}$ such that for every prefix z of w we have $\#(z,a) \geq \#(z,b) \geq \#(z,c)$, where $\#(z,x)$ counts the letters x in word z . The $a(4) = 9$ words are: $aaaa, aaaa, abaa, abaa, aabb, aabc, abac, abca, abca$. - [Alois P. Heinz](#), May 26 2012

Number of length- n restricted growth strings (RGS) $(r(1), r(2), \dots, r(n))$ such that $r(1) \leq 1$, $r(k+1) \leq r(k)$ and $r(k) = (k-1)$; for example, the 9 RGS for $n=4$ are 1010, 1012, 1201, 1210, 1212, 1230, 1231, 1232, 1234. - [Jorge Arino](#), Apr 16 2013

Number of length- n restricted growth strings (RGS) $[r(1), r(2), \dots, r(n)]$ such that $r(1)=0$, $r(k) \leq k$ and $r(k) \neq r(k-1) \implies r(k-1) = k-1$; for example, the 9 RGS for $n=4$ are 0000, 0002, 0003, 0004, 0022, 0024, 0033, 0222, 0224.

Number of (4231,5276143)-avoiding involutions in S_n . - [Alexander Burstein](#), Mar 05 2014
a(n) is the number of increasing unary-binary trees with n nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A245888](#). - [Wanda Riehl](#), Aug 02 2014

CONFIDENTIAL May 07, 2014

From [Tony Foster III](#), Jul 28 2016: (Start)
A series created using $2a(n) + a(n+1)$ has Hankel transform of $F(2n)$, offset 3, F being the Fibonacci bisection, [A001986](#) (empirical observation).
A series created using $2a(n) + 3a(n+1) + a(n+2)$ gives the Hankel transform of $\text{Sum}(k=0..n) k! \text{fibonacci}(2k)$, offset 3, [A197498](#) (empirical observation). (End)
Conjecture: $(2n)!n! \text{Sum}(k=1..n) (2k+1)a(k)^2$ is an integer for each positive integer n . - [Zhi-Wei Sun](#), Nov

The Rubey and Stump reference proves a refinement of a conjecture of René Marczinzik, which they state as: "The number of 2-Gorenstein algebras which are Nakayama algebras with n simple modules and have an oriented line as associated quiver equals the number of Motzkin paths of length n ." — [Eric M. Schmidt](#), Dec 16, 2017

Number of l -[k]-equivalence classes of Łukasiewicz paths. Łukasiewicz paths are P -equivalent iff the positions of pattern P are identical in these paths. - [Sergey Kirgizov](#), Apr 08 2018

If τ_1 and τ_2 are two distinct permutation patterns chosen from the set $\{132, 231, 312\}$, then $a(n)$ is the number of valid hook configurations of permutations of $[n+1]$ that avoid the patterns τ_1 and τ_2 . - [Colin Defant](#), Apr 28 2019

Number of permutations of length n that are sorted to the identity by a consecutive-231-avoiding stack. -

number of permutations of length n that are sorted to the identity by a consecutive-21-avoiding stack followed by a classical-21-avoiding stack. = [Colin Defant](#), Aug 29 2020

From [Helmut Prodinger](#), Dec 13 2020: (Start)

$a(n)$ is the number of paths in the first quadrant starting at $(0,0)$ and consisting of n steps from the

definition and $f(1,1), f(1,2), f(2,1), f(2,2), f(3,1), f(3,2), f(3,3)$

This step set is inspired by $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$, suggested by Emeric Deutsch around 2000.

Named by Donaghey (1977) after the Israeli-American mathematician Theodore Motzkin (1908–1970). In Sloane's "A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". - [Amirán Eldar](#), Apr 15 2021

number of Motzkin n -paths ain) is split into $A187587(n)$, number of even Motzkin n -paths, and $A343386(n)$, number of odd Motzkin n -paths. The value $A187587(n) - A343386(n)$ can be called the "shadow" of ain (see [A343773](#)). - Gennady Eremin, May 17 2021

Conjecture: If p is a prime of the form $6n+1$ (A002475), then $a(p-2)$ is divisible by p . Currently, no counterexample exists for $p < 10^7$. Personal communication from [Robert Gerbicz](#): mod such p this is equivalent to A066796 with comment: "Every A066796(n) from A066796((p-1)/2) to A066796(p-1) is divisible by prime p of form $6n+1$ ". - [Serge Batalov](#), Feb 08 2022

From [Rob Burns](#), Nov 11 2024: (Start)

The conjecture is proved in the 2017 paper by Rob Burns in the Links below. The result is contained in Tables 4 and 5 of the paper, which show that $a(p-2) \equiv 0 \pmod{p}$ when $p \equiv 1 \pmod{6}$ and $a(p-2) \equiv -1 \pmod{p}$ when $p \equiv -1 \pmod{6}$.

In fact, the 2017 paper by Burns establishes more general congruences for $a(p^k - 2)$ where $k \geq 1$.
 If $p \equiv 1 \pmod{6}$ then $a(p^k - 2) \equiv 0 \pmod{p}$ for $k \geq 1$.
 If $p \equiv -1 \pmod{6}$ then $a(p^k - 2) \equiv -1 \pmod{p}$ when k is odd and $a(p^k - 2) \equiv 0 \pmod{p}$ when k is even.
 These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper.

The 2024 paper by Nadav Kohen also proves the conjecture. Proposition 6 of the paper states that a prime p divides $a(p-2)$ if and only if $p \equiv 1 \pmod{3}$. (End)

From Peter Rala Feb 10 2022. (Start)

Conjectures:

(1) For prime $p \equiv 1 \pmod 6$ and $n, r \geq 1$, $a(n+p^r-2) \equiv -A085717(n-1) \pmod p$, where we take $A085717(0)$

(2) For prime $p \equiv 5 \pmod{6}$ and $n \geq 1$, $a(n \cdot p - 2) \equiv -A05773(n) \pmod{p}$.
 (3) For prime $p \geq 3$ and $k \geq 1$, $a(n + p^k) \equiv a(n) \pmod{p}$ for $0 \leq n \leq (p^k - 3)$.

(4) For prime $p \geq 5$ and $k \geq 2$, $a(n + p^k) \equiv a(n) \pmod{p^2}$ for $0 \leq n \leq (p^k(k-1) - 3)$. (End)

The Hankel transform of this sequence with $a(0)$ omitted gives the period-6 sequence $[1, 0, -1, -1, 0, 1, \dots]$ which is [A818892](#) with its first term omitted, while the Hankel transform of the current sequence is

the all-ones sequence [A000012](#), and also it is the unique sequence with this property which is similar to the unique Mankel transform property of the Catalan numbers. - [Michael Somos](#), Apr 17 2022

Product_{j=1..n} Sum_{i=1..j} x_i. E.g.: a(4) = 9: 3x1^2+2x2^2, 4x1^2+2x2+3x3, 2x1^2+2x2+3x4, x1^2+2x3^2, x1^2+2x3+3x4, 2x1x2+x2^2+3x3, x1x2+2^2+3x4, x1x2+2x3+3^2, x1x2+2x3+3x4. — [Flab Baser](#), Dec 20 2024

The Hankel matrix corresponding to this sequence factorizes as $T \cdot \text{Transposon}(T)$. Where T is the lower

unitriangular array [A064189](#) (row reverse of the Motzkin triangle [A026300](#)). Hence the Hankel determinants are all equal to 1 as stated above by Michael Somos. - [Peter Bala](#), Nov 23 2025

Number of 2-stack-sortable permutations of $\{1, 2, \dots, n+1\}$ ending with 1.
 Number of standard composition tableaux of width at most 2 on n cells. (End)

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ▶ ↺

ruins	August 2026	15
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August 2020 15 /

*n*th Motzkin counts:

Number of 4321-, {3412, 2413}, {3412, 3142}- and 3412-avoiding involutions in S_n .
 Number of sequences of length $n-1$ consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any 2 consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003
 From [David Callan](#), Jul 15 2004: (Start)
 Also number of Motzkin n -paths: paths from $(0,0)$ to $(n,0)$ in an $n \times n$ grid using only steps $U = (1,1)$, $F = (1,0)$ and $D = (1,-1)$.

Number of Dyck n -paths with no UUU . (Given such a Dyck n -path, change each UUD to U , then change each remaining UD to F . This is a bijection to Motzkin n -paths. Example with $n=5$: $UUDUDUUDDD \rightarrow UFUUDDD$.)

[illegible]

a(n) = number of Dyck n-paths all of whose valleys have even x-coordinate (when path starts at origin). For example, $T(4,2)=3$ counts UDUUDD, UDUUDDU, UUUUUUD. Given such a path, split it into n subpaths of length 2 and transform $UD \rightarrow U$, $DD \rightarrow D$, $UD \rightarrow F$ (there will be no DUs for that would entail a valley with odd x-coordinate). This is a bijection to Motzkin n-paths. — David Collin, Jun 87 2895

Also the number of standard Young tableaux of height = k = 3 - Mike Zabrocki, Mar 24 2007

a(n) is the number of RNA shapes of size 2n-2. RNA Shapes are essentially Dyck words without "directly nested" motifs of the form A[B]B[C], for A, B and C Dyck words. The first RNA Shapes are []; [1]; [[][]]; [[[]]] ; [[[]][[]]]; [[[]][[]]][[]]; [[[]][[]]][[]][[]]; [[[]][[]]][[]][[]][[]]; ...
The sequence is self-generated from top row A going to the left starting (1,1) and bottom row B, the same sequence but starting (0,1) and going to the right. Take dot product of A and B and add the result to n-th term of A to get the (n+1)-th term of A. Example: a(5) = 21 as follows: Take dot product of A = (9, 4, 2, 1, 0) and B = (0, 1, 2, 3, 4).

2088
Equals $A055773 / A05773$ shifted (i.e., $(1,2,5,13,35,96,\dots) / (1,1,2,5,13,35,96,\dots)$). - Gary W. Adamson,
Dec 21 2008
Starting with offset 1 = iterates of $M * [1,1,0,0,\dots]$, where M = a tridiagonal matrix with $[0,1,1,\dots]$
in the main diagonal and $[1,1,1,\dots]$ in the super and subdiagonals. - Gary W. Adamson, Jan 07 2009.

$n!$ is the number of involutions of $\{1, 2, \dots, n\}$ having genus g . The genus $g(p)$ of a permutation p of $\{1, 2, \dots, n\}$ is defined by $g(p) = (1/2)(n-1 - (p - \text{cpl}(p)))$, where p^* is the inverse permutation of p , $c = 234\dots n-1 = \{1, 2, \dots, n\}$, and $\text{cpl}(p)$ is the number of cycles of the permutation p . Example: $a(4) = 9$; indeed, $a(4) = 9$ because there are 9 involutions of $\{1, 2, 3, 4\}$ having genus 0. Note that a permutation p of $\{1, 2, \dots, n\}$ has genus 0 if and only if the cycle decomposition of p gives an undirected partition of $\{1, 2, \dots, n\}$ and each cycle of p is increasing (see Lemma 2.1 of the Dulucq-Simon reference). [Also, redundantly, for $p = 3412$ in [13] [24] we have $\text{cpl}(p) = 2341 + 3412 + 2123 = 1432$ and so $g(p) = (1/2)(4 + 1 - 2) = 1$.]

Let $w(i, j, n)$ denote walks in $\mathbb{N}^2$ which satisfy the multivariate recurrence $w(i, j, n) = w(i, j+1, n-1) + w(i-1, j, n-1) + w(i+1, j-1, n-1)$ with boundary conditions $w(0, 0, 0) = 1$ and $w(i, j, n) = 0$ if i or j or $n < 0$. Then $a(n) = \sum_{i=0..n} \sum_{j=0..n} w(i, j, n)$ is the number of such walks of length n . -

$\alpha(N-1)$ tends to 3.0 as $N \rightarrow \infty$: $1+2\cos(2\pi/N)$ relating to longest odd N regular polygon diagonals, by way of example, $N=7$: Using the triangular generator [cf. comment of Jan 07 2009], for polygon $N=7$, we extract an $(N-1)/2 = 3 \times 3$ matrix, $\{0,1\}$, $\{0,1,1; 1,1,1; 0,1,1\}$ with an e-val of $2.4697\dots$; the longest Heptagon diagonal with length 3 . As N tends to infinity, the diagonal lengths tend to 3.0 , the convergence of the sequence. — Gary W. Adamson, Jun 08 2011

Number of $(n-1)$ -length permutations avoiding the pattern 132 and the dotted pattern 231d{01}. — Jean-Luc Baril, Mar 07 2012

Number of n -length words w over alphabet $\{a,b,c\}$ such that for every prefix z of w we have $\#(z,a) \geq \#(z,b) \geq \#(z,c)$, where $\#(z,x)$ counts the letters x in word z . The $a(4) = 9$ words are: $aaaa, aaaa, abaa, abaa, aabb, aabc, aabc, abaca, abaca$. - [Alois P. Heinz](#), May 26 2012

Number of length- n restricted growth strings (RGS) $(r(1), r(2), \dots, r(n))$ such that $r(1)=1$, $r(k) \leq k$, and $r(k) = (k-1)$; for example, the 9 RGS for $n=4$ are 1010, 1012, 1201, 1210, 1212, 1230, 1231, 1232, 1234. - [Jorge Arregui](#), Apr 16 2013

Number of length- n restricted growth strings (RGS) $[r(1), r(2), \dots, r(n)]$ such that $r(1)=0$, $r(k) \leq k$ and $r(k) \neq r(k-1) \implies r(k-1) = k-1$; for example, the 9 RGS for $n=4$ are 0000, 0002, 0003, 0004, 0022, 0024, 0033, 0222, 0224.

a(n) is the number of increasing unary-binary trees with n nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A245888](#). - [Manda Bickel](#), Aug 07 2014

From [Tony Foster III](#), Jul 28 2016: (Start)
 A series created using $2a(n) + a(n+1)$ has Hankel transform of $F(2n)$, offset 3, F being the Fibonacci bisection, [A001396](#) (empirical observation).
 A series created using $2a(n) + 3a(n+1) + a(n+2)$ gives the Hankel transform of $\text{Sum}(k=0..n) k! \text{fibonacci}(24k)$, offset 3, [A197549](#) (empirical observation). (End)
 Conjecture: $(2n)(n+1)k! \text{fib}(2n+1+k) = (2k+1)a(k)^2$ is an integer for each positive integer n . - [Zhi-Wei Sun](#), Nov

The Rubey and Stump reference proves a refinement of a conjecture of René Marczinzik, which they state as:
 "The number of 2-Gorenstein algebras which are Nakayama algebras with n simple modules and have an oriented line as associated quiver equals the number of Motzkin paths of length n ." – [Eric M. Schmidt](#), Dec 16, 2017.

Number of U_k -equivalence classes of lukasiewicz paths. lukasiewicz paths are P -equivalent iff the positions of pattern P are identical in these paths. - [Sergey Kirgizov](#), Apr 88 2018
If τ_1 and τ_2 are two distinct permutation patterns chosen from the set $\{132, 312, 312\}$, then $a(n)$ is the number of valid hook configurations of permutations of $[n+1]$ that avoid the patterns τ_1 and τ_2 . - [Colin Defant](#), Apr 28 2019

Number of permutations of length n that are sorted to the identity by a consecutive-321-avoiding stack followed by a classical-21-avoiding stack. - [Colin Defant](#), Aug 29 2020
From [Helmut Prodinger](#), Dec 13 2020: (Start)
 $a(n)$ is the number of paths in the first quadrant starting at $(0,0)$ and consisting of n steps from the

For example, denoting $U=(1,1)$, $D=(1,-1)$, $D_{-j}=(1,-j)$ for $j \geq 2$, $a(4)$ counts $UUUU$, $UUUD$, $UUUD_2$, $UUUD_3$, $UUUD_4$, $UUUD_5$, $UUUD_6$, $UUUD_7$, $UUUD_8$, $UUUD_9$, $UUUD_{10}$.

This step set is inspired by $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$, suggested by Emeric Deutsch around 2000. See Prodinger link that contains a bijection to Motzkin paths. (End)
 Named by Donaghey (1977) after the Israeli-American mathematician Theodore Motzkin (1900-1970). In Sloane's "A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". - Amir Eldar, Jan 15, 2024

Number of Motzkin n -paths $a(n)$ is split into [A107587](#)(n), number of even Motzkin n -paths, and [A343386](#)(n), number of odd Motzkin n -paths. The value [A107587](#)(n) - [A343386](#)(n) can be called the "shadow" of $a(n)$ (see [A343773](#)). - Gennady Eremin, May 17 2021

Conjecture: If p is a prime of the form $6n+1$ ([A002476](#)), then $a(p-2)$ is divisible by p . Currently, no counterexample exists for $p < 10^7$. Personal communication from [Robert Gerbicz](#): mod such p this is equivalent to [A056796](#) with comment: "Every [A056796\(n\) from \[A056796\]\(#\)\(\$\(p-1\)/2\$ \) to \[A056796\]\(#\)\(\$p-1\$ \) is divisible by prime \$p\$ of form \$6n+1\$ ". - \[Serge Batalov\]\(#\), Feb 08 2022](#)

The conjecture is proved in the 2017 paper by Rob Burns in the Links below. The result is contained in Tables 4 and 5 of the paper, which show that $a(p-2) \equiv 0 \pmod{p}$ when $p \equiv 1 \pmod{6}$ and $a(p-2) \equiv -1 \pmod{p}$ when $p \equiv -1 \pmod{6}$.

In fact, the 2017 paper by Burns establishes more general congruences for $a(p^k - 2)$ where $k \geq 1$.
 If $p \equiv 1 \pmod{6}$ then $a(p^k - 2) \equiv 0 \pmod{p}$ for $k \geq 1$.
 If $p \equiv -1 \pmod{6}$ then $a(p^k - 2) \equiv -1 \pmod{p}$ when k is odd and $a(p^k - 2) \equiv 0 \pmod{p}$ when k is even.
 These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper.

The 2024 paper by Nadav Kohen also proves the conjecture, Proposition 6 of the paper states that a prime p divides $a(p-2)$ if and only if $p \equiv 1 \pmod{3}$. (End)
From [Peter Bala](#), Feb 10 2022: (Start)
[Proof-idea](#)

conjectures:

- (1) For prime $p \equiv 1 \pmod{6}$ and $n, r \geq 1$, $a(n+p^r - 2) \equiv -A057173(n-1) \pmod{p}$, where we take $A057173(0) = 0$ to match Batylov's conjecture above.
- (2) For prime $p \equiv 5 \pmod{6}$ and $n \geq 1$, $a(np - 2) \equiv -A057733(n) \pmod{p}$.
- (3) For prime $p \equiv 3 \pmod{4}$ and $k \geq 1$, $a(n + p^k) \equiv a(n) \pmod{p}$ for $0 \leq n \leq (p^k - 3)$.
- (4) For prime $p \equiv 5$ and $k \geq 2$, $a(n + p^k) \equiv a(n) \pmod{p^2}$ for $0 \leq n \leq (p^{k-1} - 3)$. (End)

The Hankel transform of the sequence with a(8) omitted gives the period-6 sequence [1, 0, -1, -1, 0, 1, ...] which is [A018892](#) with its first term omitted, while the Hankel transform of the current sequence is the all-ones sequence [A000012](#), and also it is the unique sequence with this property which is similar to the unique Hankel transform property of the Catalan numbers. - [Michael Somos](#), Apr 17 2022

The number of terms in which the exponent of any variable x_i is not greater than 2 in the expression of $\text{Product}_{i=1}^n (1+x_i+\text{sum}_{j=1}^i x_j)$ is $a(n)$. e.g. $\text{sum}_{j=1}^1 x_j = x_1$, $a(1) = 0$; $\text{sum}_{j=1}^2 x_j = x_1 + x_2$, $a(2) = 2$; $\text{sum}_{j=1}^3 x_j = x_1 + x_2 + x_3$, $a(3) = 4$; $\text{sum}_{j=1}^4 x_j = x_1 + x_2 + x_3 + x_4$, $a(4) = 8$; $\text{sum}_{j=1}^5 x_j = x_1 + x_2 + x_3 + x_4 + x_5$, $a(5) = 16$; $\text{sum}_{j=1}^6 x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$, $a(6) = 32$; $\text{sum}_{j=1}^7 x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7$, $a(7) = 64$; $\text{sum}_{j=1}^8 x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$, $a(8) = 128$; $\text{sum}_{j=1}^9 x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9$, $a(9) = 256$; $\text{sum}_{j=1}^{10} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10}$, $a(10) = 512$; $\text{sum}_{j=1}^{11} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11}$, $a(11) = 1024$; $\text{sum}_{j=1}^{12} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12}$, $a(12) = 2048$; $\text{sum}_{j=1}^{13} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13}$, $a(13) = 4096$; $\text{sum}_{j=1}^{14} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14}$, $a(14) = 8192$; $\text{sum}_{j=1}^{15} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15}$, $a(15) = 16384$; $\text{sum}_{j=1}^{16} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16}$, $a(16) = 32768$; $\text{sum}_{j=1}^{17} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17}$, $a(17) = 65536$; $\text{sum}_{j=1}^{18} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18}$, $a(18) = 131072$; $\text{sum}_{j=1}^{19} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19}$, $a(19) = 262144$; $\text{sum}_{j=1}^{20} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20}$, $a(20) = 524288$; $\text{sum}_{j=1}^{21} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21}$, $a(21) = 1048576$; $\text{sum}_{j=1}^{22} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22}$, $a(22) = 2097152$; $\text{sum}_{j=1}^{23} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23}$, $a(23) = 4194304$; $\text{sum}_{j=1}^{24} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24}$, $a(24) = 8388608$; $\text{sum}_{j=1}^{25} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24} + x_{25}$, $a(25) = 16777216$; $\text{sum}_{j=1}^{26} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26}$, $a(26) = 33554432$; $\text{sum}_{j=1}^{27} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27}$, $a(27) = 67108864$; $\text{sum}_{j=1}^{28} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27} + x_{28}$, $a(28) = 134217728$; $\text{sum}_{j=1}^{29} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27} + x_{28} + x_{29}$, $a(29) = 268435456$; $\text{sum}_{j=1}^{30} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27} + x_{28} + x_{29} + x_{30}$, $a(30) = 536870912$; $\text{sum}_{j=1}^{31} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27} + x_{28} + x_{29} + x_{30} + x_{31}$, $a(31) = 1073741824$; $\text{sum}_{j=1}^{32} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} + x_{21} + x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27} + x_{28} + x_{29} + x_{30} + x_{31} + x_{32}$, $a(32) = 2147483648$; $\text{sum}_{j=1}^{33} x_j = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x$

The Hankel matrix corresponding to this sequence factorizes as $T * \text{Transpose}[T]$, where T is the lower unitriangular array [A054189](#) (row reverse of the Motzkin triangle [A026308](#)). Hence the Hankel determinants

From **Michael Luo** Jul 29 2026: (Start)
Number of 2-stack-sortable permutations of $\{1, 2, \dots, n+1\}$ ending with 1.
Number of standard composition tableaux of width at most 2 on n cells. (End)

From [Michael Luo](#), Jul 29 2026: (Start)
Number of 2-stack-sortable permutations
of $\{1, 2, \dots, n + 1\}$ ending with 1.

Acknowledgements

I thank

- my mentor Ryota Inagaki
- my tutor AnaMaría Perez
- Tanya Khovanova, David Jerison, Jonathan Bloom
- TAs Jaeho Lee, Frederick Song, Celine Zhang
- Timothy Chen, Xindi Luo, and Sowmya Sankaran
- Colin Defant for help formulating the project
- Jerry Zhang
- MIT and CEE
- my sponsor Joseph Mandarino

s^2 -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

s^2 -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

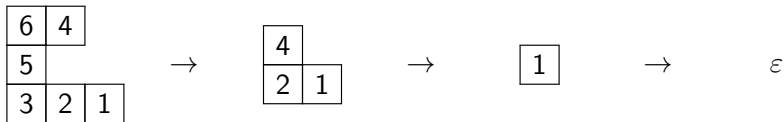
s^t -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width $\leq t$

s^2 -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

s^t -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width $\leq t$

Applications of s correspond to cutting off columns:

4652130 $\xrightarrow{s}$ 4120356 $\xrightarrow{s}$ 1023456 $\xrightarrow{s}$ 0123456



Power $p(c)$ of a cell:

10	9	5	
8	7		
6			
4	3	2	1

Theorem (Zhang 2026)

of words in S'_n corresponding to T is

$$\prod_c p(c),$$

where product taken over $c \in T$ not in first column.

Theorem (Zhang 2026)

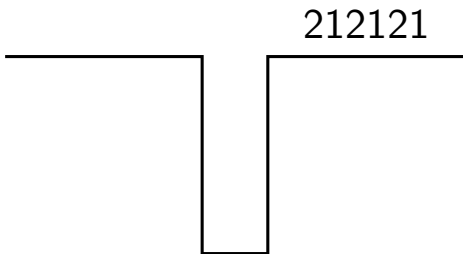
The number of s^t -sortable permutations in S'_n is computable in polynomial time in n .

Theorem (Zhang 2026)

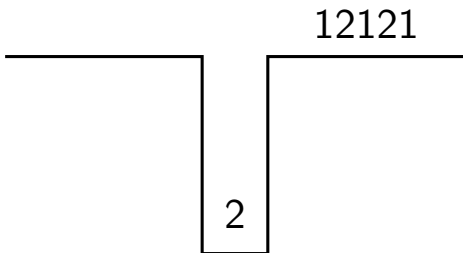
Let $f(n)$ be the average number of iterations needed to sort $\pi \in S_n$. Then $\lim_{n \rightarrow \infty} \frac{f(n)}{n}$ exists.

s_m allows m identical penguins in hole.

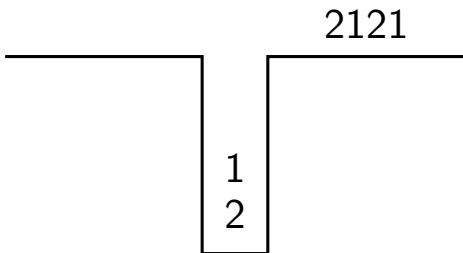
s_m allows m identical penguins in hole. s_2 :



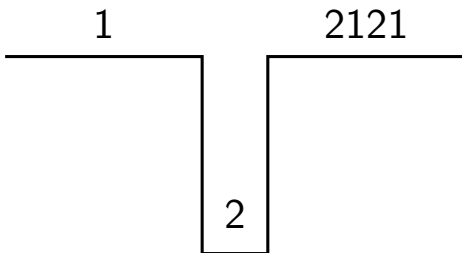
s_m allows m identical penguins in hole. s_2 :



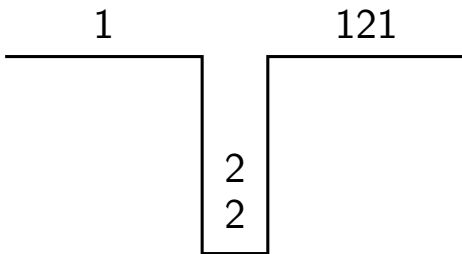
s_m allows m identical penguins in hole. s_2 :



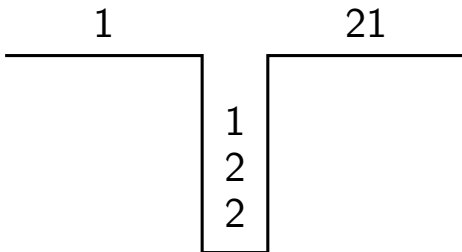
s_m allows m identical penguins in hole. s_2 :



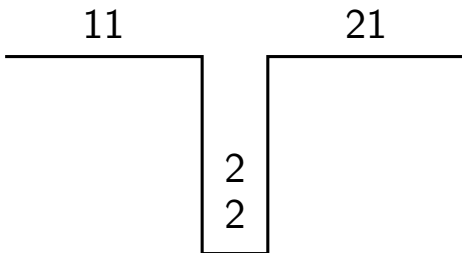
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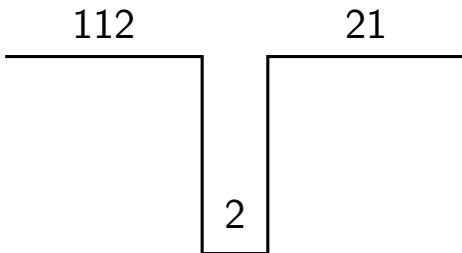
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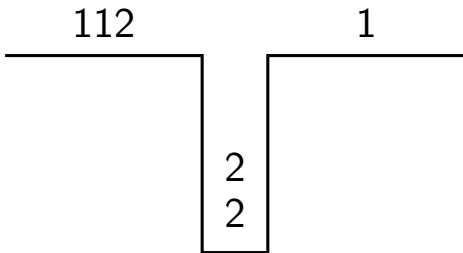
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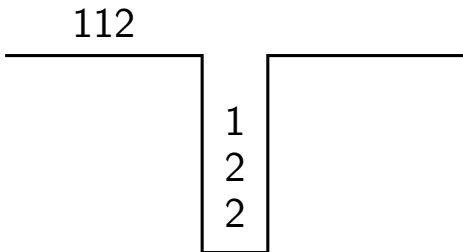
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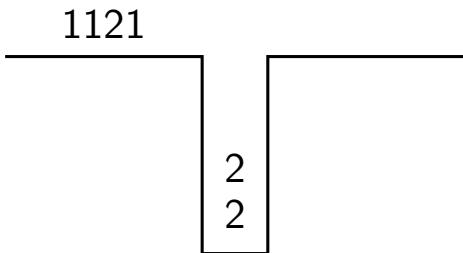
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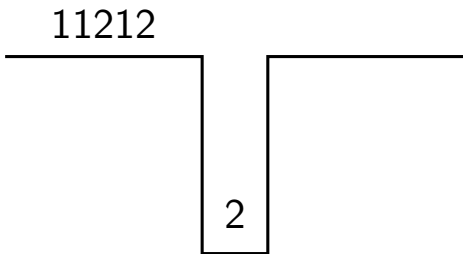
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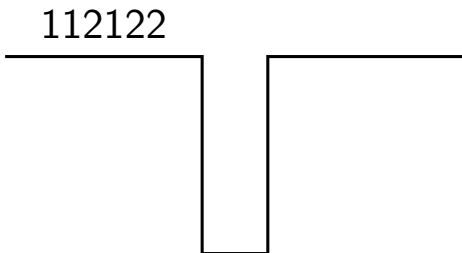
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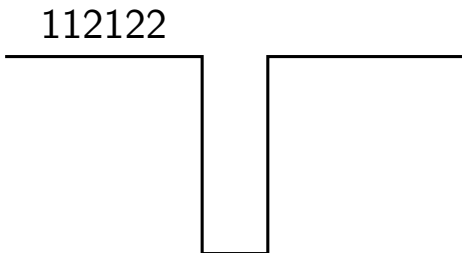
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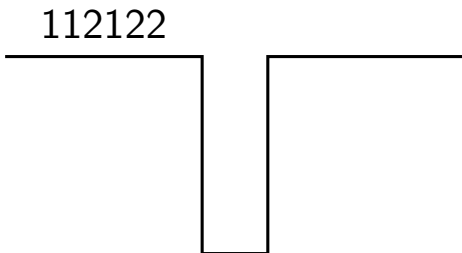


s_m allows m identical penguins in hole. s_2 :



s_7 “more powerful” than s_6 : every s_6 -sortable word s_7 -sortable.

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s_7 “more powerful” than s_6 : every s_6 -sortable word s_7 -sortable. But:

Theorem (L. 2026)

For all $m \neq m'$, there are words w such that w takes much longer to sort with s_m than $s_{m'}$.